Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Dept. of Electrical and Computer Engineering

Course Code: ECE-2112

Course Title: Digital Techniques Sessional

Lab Report 02

Simplification of Boolean Expressions and Verification Using Logic Circuit Simulation

Date of Submission: 16 August, 2026

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Session: 2024-2025

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Task 01: Simplification and Verification of Boolean Expression
$F(A, B, C) = A'BC + AB'C + ABC' + ABC$

Objective

To reduce a three-variable sum-of-minterms expression to its minimal sum-of-products form, to confirm the reduction using a Karnaugh map, and to verify that the original and simplified circuits are functionally identical by implementing both in Logisim.

Algebraic Simplification

$$
\begin{aligned}
F(A, B, C) &= A'BC + AB'C + ABC' + ABC \\
&= A'BC + ABC + AB'C + ABC + ABC' + ABC \\
&= BC(A' + A) + AC(B' + B) + AB(C' + C) \\
&= BC + AC + AB \\
&= AB + BC + AC
\end{aligned}
$$

Truth Table

A	***B***	***C***	***F = AB + BC + AC***
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Table 01: Truth table for ***$F(A, B, C) = AB + BC + AC$***

Karnaugh Map (K-Map)

A\BC	00	01	11	10
0	0	0	1	0
1	0	1	1	1

Table 02: K-map for ***$F(A, B, C) = A'BC + AB'C + ABC' + ABC$***

Three groups of two adjacent 1-cells are formed: the pair at $BC = 11$ (both rows) gives BC; the pair at $A = 1$, $C = 1$ ($BC = 01, 11$) gives AC; and the pair at $A = 1$, $B = 1$ ($BC = 11, 10$) gives AB. Combining the three groups yields the minimized expression $F = AB + AC + BC$, matching the algebraic result.

Circuit Diagram

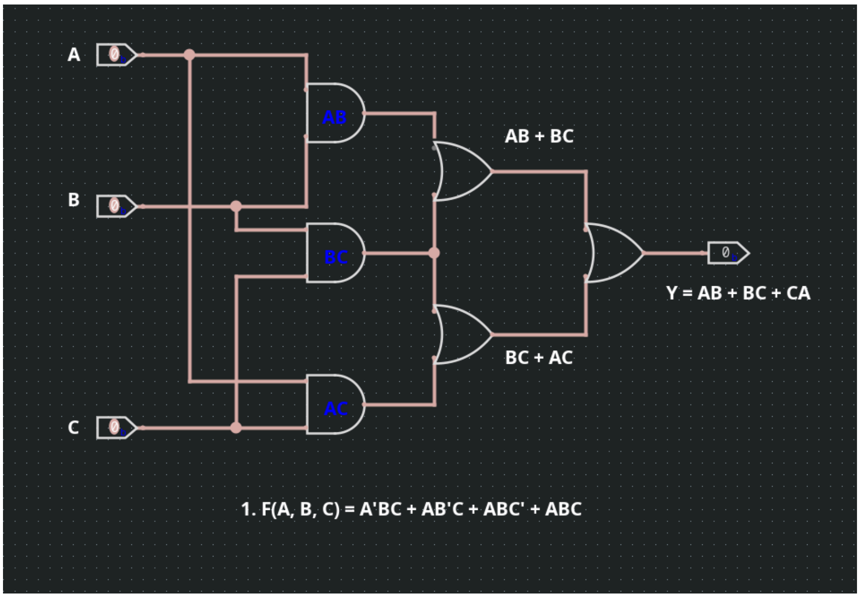

Figure 01: Simplified Circuit Diagram for *F(A, B, C) = A′BC + AB′C + ABC′ + ABC*

Discussion

The original circuit uses four 3-input AND gates, inverters, and an OR gate. After simplification, the circuit becomes AB + AC + BC, requiring only three 2-input AND gates and an OR gate. Logisim simulation showed that both circuits produced identical outputs, confirming the simplification.

Conclusion

The expression F = A′BC + AB′C + ABC′ + ABC was simplified to F = AB + AC + BC using Boolean algebra and a Karnaugh map. Simulation verified that the simplified circuit gives the same output as the original for all input combinations.

Task 02: Simplification and Verification of the Boolean Expression $F(A, B, C) = A(A+B)(A+B+C)$

Objective

To reduce a three-variable product-of-sums-type expression to its simplest form, to con-firm the reduction using a Karnaugh map, and to verify the equivalence of the original and simplified circuits in Logisim.

Algebraic Simplification

$$
\begin{aligned}
F(A, B, C) &= A(A + B)(A + B + C) \\
&= (A + AB)(A + B + C) \\
&= A(1 + B)(A + B + C) \\
&= A(A + B + C) \\
&= A + AB + AC \\
&= A(1 + B + C) \\
&= A
\end{aligned}
$$

Truth Table

A	*B*	*C*	*F = A*
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Table 03: Truth table for $F(A, B, C) = A$

Karnaugh Map

A\BC	**00**	**01**	**11**	**10**
0	0	0	0	0
1	1	1	1	1

Table 04: K-map for $F(A, B, C) = A$

All four cells in the $A = 1$ row are 1, while the entire $A = 0$ row is 0. This forms a single group of four adjacent cells covering the whole row, which eliminates both B and C and leaves only $F = A$, exactly matching the algebraic simplification

Circuit Diagram

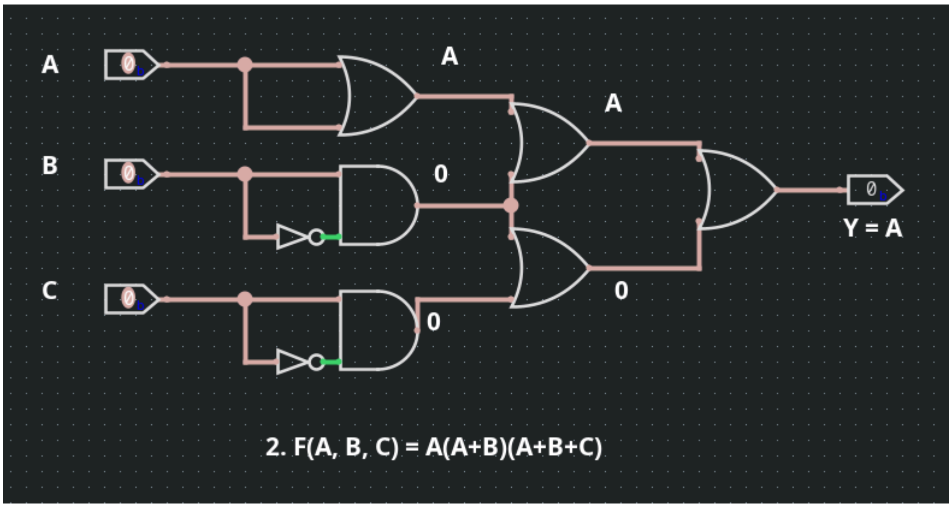

Figure 02: Simplified Circuit Diagram for F(A, B, C) = A(A+B)(A+B+C)

Discussion

The original circuit implements F = A(A + B)(A + B + C) using OR and AND gates. Applying the absorption law simplifies the expression to F = A, eliminating all logic gates. Logisim simulation confirmed that the output always equals A, regardless of B and C.

Conclusion

The expression F = A(A + B)(A + B + C) was simplified to F = A using the absorption law. Karnaugh map analysis and Logisim simulation verified that the simplified circuit produces the same output as the original for all input combinations.

Task 03: Simplification and Verification of the Boolean Expression *F(A, B, C) = (A + (BC)')'(AB + ABC)*

Objective

To reduce a mixed AND-OR-complement expression using De Morgan's theorem and Boolean algebra, to confirm the result with a Karnaugh map, and to verify using Logisim that the simplified circuit behaves as a constant.

Algebraic Simplification

$F(A, B, C) = (A + (BC)')'(AB + ABC)$
$= (A + B' + C')'\ AB\ (1+C)$
$= AB\ (A'BC)$
$= (AA')BC$
$= 0$

Truth Table

A	*B*	*C*	*F = 0*
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Table 05: Truth table for *F(A, B, C) = 0*

Karnaugh Map

A\BC	00	01	11	10
0	0	0	0	0
0	0	0	0	0

Table 06: K-map for *F(A, B, C) = 0*

Every cell of the map is 0, so there are no 1-cells to group. A Karnaugh map with no populated cells represents the constant function F = 0, confirming the algebraic result.

Circuit Diagram

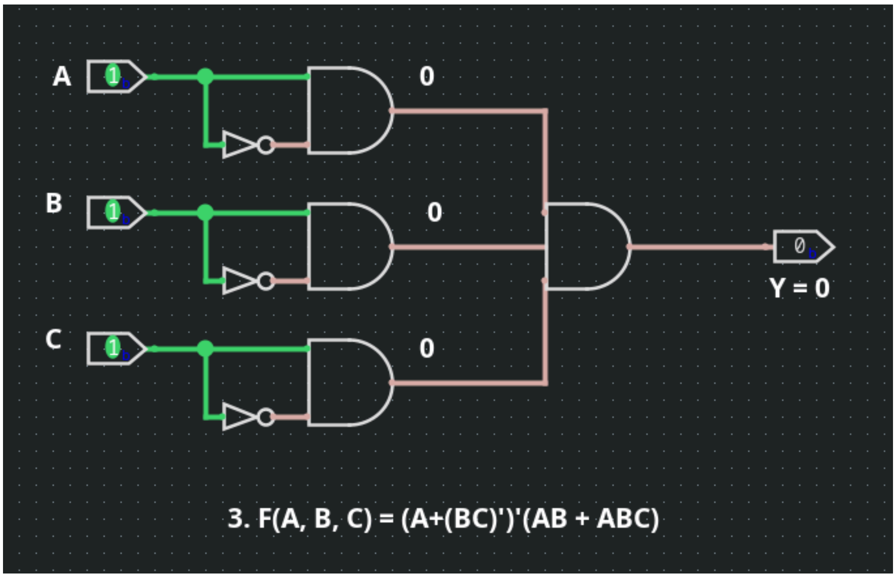

Figure 03: Simplified Circuit Diagram for *F(A, B, C) = (A+(BC)′)′(AB+ABC)*

Discussion

Using De Morgan's theorem and the absorption law, the expression simplifies to 0 because it contains the term $A \cdot A' = 0$. Logisim simulation confirmed that the output remains 0 for all input combinations of A, B, and C.

Conclusion

The expression $F = (A + (BC)')'(AB + ABC)$ simplifies to 0. Boolean algebra, Karnaugh map analysis, and Logisim simulation all verified that the output is always logic 0.

Task 04: Simplification and Verification of the Boolean Expression ***F(A, B) = (B′(A + B) + (A + B)(A + B′))B′***

Objective

To reduce a two-variable expression built from nested OR and AND terms to its minimal form, to confirm the reduction using a Karnaugh map, and to verify the equivalence of the original and simplified circuits in Logisim.

Algebraic Simplification

F(A, B) = (B’(A+B) + (A+B)(A+B’))B’
= B’(AB’ + BB’ + AA + AB’ + AB + BB’)
= B’(AB’ + A + AB’ + AB)
= AB’(B’ + 1 + B’ + B)
= AB’

Truth Table

A	B	F(A, B) = AB’
0	0	0
0	1	0
1	0	1
1	1	0

Table 07: Truth table for ***F(A, B) = AB’***

Karnaugh Map

A\B	0	1
0	0	0
1	1	0

Table 08: K-map for ***F(A, B)’= (B’(A + B) + (A + B)(A + B’))B’***

Only a single cell, at *A* = 1, *B* = 0, is populated. A lone 1-cell cannot be combined with any adjacent cell, so it is read directly as the product of its coordinate literals, giving *F* = *AB*′, matching the algebraic result.

Circuit Diagram

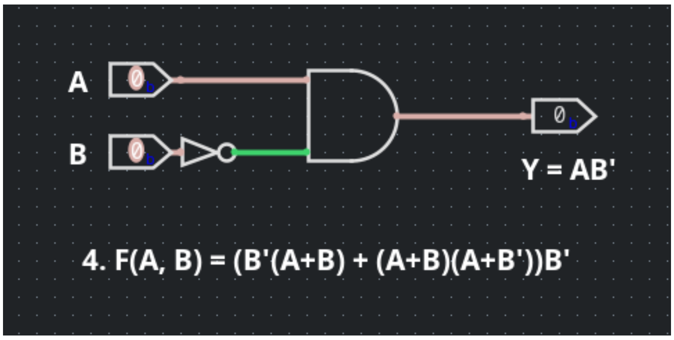

Figure 04: Simplified Circuit Diagram for *F(A, B) = (B′(A+B) + (A+B)(A+B′))B′*

Discussion

Using Boolean algebra, the expression simplifies to F = AB′. The simplified circuit requires only an inverter on B and a 2-input AND gate. Logisim simulation confirmed that both the original and simplified circuits produce identical outputs for all input combinations.

Conclusion

The expression F = (B′(A + B) + (A + B)(A + B′))B′ was simplified to F = AB′. The result was verified using Boolean algebra, a Karnaugh map, and Logisim simulation.

References

[1] M. M. Mano and M. D. Ciletti, Digital Design, 5th ed. Upper Saddle River, NJ, USA: Pearson, 2013.

[2] R. J. Tocci, N. S. Widmer, and G. L. Moss, Digital Systems: Principles and Applications, 11th ed. Upper Saddle River, NJ, USA: Pearson, 2011.